

Victor D. Masters

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Professional Summary

Control Systems Programmer with over 20 years of experience architecting safety-critical automation, SCADA, and data acquisition (DAQ) systems for industrial and oil & gas environments. Proven expertise in developing machine control software using modern object-oriented languages (Python/C), authoring detailed functional specifications, and delivering comprehensive control system documentation. Adept at collaborating with automation engineers and technicians to test, troubleshoot, and debug complex hydraulic, electrical, and pneumatic systems while supporting accurate project time and effort estimates. Committed to continuous improvement and cross-functional leadership to drive operational efficiency, system reliability, and production optimization.

Professional Experience

Control Systems & Embedded Development Lead — Oil & Gas / Embedded Systems Sector (Austintown, OH) | January 2004 – Present

- Architect and program safety-critical machine control systems and custom SCADA/DAQ/telemetry solutions, directly delivering \$23 million in production increases.
- Develop and maintain embedded hardware/software control applications using modern object-oriented languages (Python/C) across PIC, PLC, ARM, and ESP32 platforms.
- Author detailed control system functional specifications and technical documentation to guide automation engineers, technicians, and end customers through system deployment.
- Test, troubleshoot, and debug complex control systems, reducing annual operational downtime from 30–45 days to under 10 days through rigorous validation and standardized maintenance protocols.
- Integrate hydraulic, electrical, and pneumatic systems into automated control architectures, ensuring seamless process instrumentation and full-cycle production supervision.
- Support project time and effort estimates for new automation applications while leading cross-functional teams in system integration, quality assurance, and continuous improvement initiatives.
- Develop and deliver comprehensive training programs for technical staff, enhancing team capability in control system programming, system debugging, and operational safety standards.

Technical Skills

Control System Architecture: SCADA & DAQ Development, Safety-Critical Programming, Machine Control Systems, PLC & Embedded Systems (PIC/ARM/ESP32) Programming & OOP: Python, C/C++, Object-Oriented Programming, Software Development, System Integration Engineering & Documentation: Functional Specifications, Control System Documentation, Process Instrumentation, Hydraulic/Electrical/Pneumatic Systems Operations & Collaboration: System Debugging & Troubleshooting, Project Estimation, Cross-Functional Collaboration, Continuous Improvement, Telemetry

Key Achievements

- Delivered \$23 million in production increases by architecting and programming safety-critical machine control systems and custom SCADA/DAQ/telemetry solutions.
- Reduced annual operational downtime from 30–45 days to under 10 days through rigorous system debugging, troubleshooting, and implementation of standardized maintenance protocols.
- Authored comprehensive control system functional specifications and technical documentation, enabling seamless cross-functional collaboration with automation engineers and technicians.
- Developed and maintained embedded hardware/software control applications using modern object-oriented languages (Python/C) across PIC, PLC, ARM, and ESP32 platforms.

Education

Bachelor's Degree in Technical Discipline

Additional Skills

U.S. Export Compliance Awareness | Drug-Free Workplace Compliance | Agile Project Management | Quality Assurance | Production Optimization